

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)

Course Name: Applied Mathematics-1
Examination: 1st Periodical Examinations
Branch: M.E./ECE/CSE

Course Code: MA-111
B.Tech Scheme: NEP
Semester: 1st

Question Paper for 1st Periodical Examination (OCT 2025)

Time: One and half Hours

Max Marks: 30

Note: Attempt all questions

CO No.	Course Outcomes
CO.1	Examine linear simultaneous equations for consistency and applications
CO.2	Apply concept of linear algebra to convert quadratic form to canonical forms
CO.3	Solve problems on improper integral & Inspect the function of two variables for optimization
CO.4	Evaluate double and triple integrals in various coordinates & their application in engineering
CO.5	Test for the analyticity of complex functions
CO.6	Solve problems on complex integration

Q.No.	Questions	Marks	CO	BL
1.a	By using elementary transformation find the inverse of matrix $A = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix}$	1,1	CO1	L1
1.b	Define Hermitian and Skew Hermitian matrix	1	CO1	L1
1.c	If A is orthogonal matrix prove that $ A = \pm 1$	1	CO1	L1
1.d	Define Cayley-Hamilton theorem	1	CO1	L2
2.	Reduce the following matrix to echelon form and find its Rank: $A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$	5	CO1	L3
3.	Solve with the help of Gauss Jordan Method: $x+y+z=3$, $x+2y+3z=4$, $x+4y+9z=6$	5	CO2	L3
4.	Find the Eigen values and the corresponding Eigen vectors of the matrix. $\begin{bmatrix} -2 & 2 & 3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$	5	CO2	L3
5.	(a) Reduce the matrix into diagonal form $A = \begin{bmatrix} -1 & 2 & -2 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix}$ (b) Find the value of k for which the system of linear equations $x+y+z=1$, $2x+y+4z=k$, $4x+y+10z=k^2$ have a solution, and solve.	5,5	CO2, CO1	L5

Faculty Name: Rita Rana (MSH)